Price Elasticity of Demand

Price elasticity of demand is commonly known as elasticity of demand. This is because price is the most influential factor affecting demand. "Elasticity of demand measures the responsiveness of the quantity demanded to changes in the price".

1. **Price Elasticity of Demand:** The price elasticity of demand, commonly known as the elasticity of demand refers to the responsiveness and sensitiveness of demand for a product to the changes in its price. In other words, the price elasticity of demand is equal to

$$E_p = \frac{\textit{Proportionate change in Quantity Demanded}}{\textit{Proportionate change in Price}}$$

Numerically,

$$E_p = \frac{\Delta Q}{\Delta P} X \frac{P}{Q}$$

ΔQ = changes in demand.

ΔP = changes in price.

P = original price.

Q = original quantity.

where, $\Delta Q = Q_1 - Q_0$, $\Delta P = P_1 - P_0$, Q_1= New quantity,

Q_0= Original quantity, P_1 = New price, P_0 = Original price.

2. **Income Elasticity of Demand:** The income is also a factor that influences the demand for a product. Hence, the degree of responsiveness of a change in demand for a product due to the change in the income is known as income elasticity of demand. The formula to compute the income elasticity of demand is:

$$E_Y = \frac{\text{Proportionate change in Quantity Demand for a product}}{\text{Proportionate change in Income}}$$

For most of the goods, the income elasticity of demand is greater than one indicating that with the change in income the demand will also change and that too in the same direction, i.e. more income means more demand and vice-versa.

3. **Cross Elasticity of Demand:** The cross elasticity of demand refers to the percentage change in quantity demanded for one commodity as a result of a small change in the price of another commodity. This type of elasticity usually arises in the case of the interrelated goods such as substitutes and complementary goods. The cross elasticity of demand for goods X and Y can be expressed as:

$$E_c = \frac{\textit{Proportionate change in demand of Commodity X}}{\textit{Proportionate change in price of Commodity Y}}$$

4. **Advertising Elasticity of Demand:** The responsiveness of the change in demand due to the change in advertising or other promotional expenses, is known as advertising elasticity of demand. It can be expressed as:

$$E_a = \frac{\textit{Proportionate change in Demand}}{\textit{Proportionate change in Advertising Expenditure}}$$

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